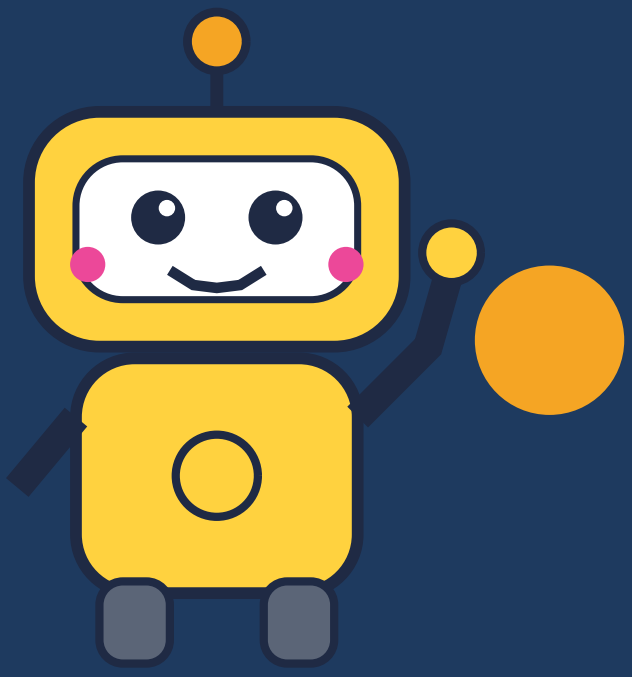
SAMPLE PAPERS

Think Like a Coder

Sample examination papers, Class 8

Half-yearly and annual papers
50 marks, 2 hours, with blueprints and marking schemes
Practical stations and project assessment



For CBSE's Computational Thinking and AI curriculum, 2026-27

Goes with Think Like a Coder, Class 8 · Photocopiable for school use

SAMPLE EDITION · DRAFT

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About these papers

These are two sample examination papers for Class 8: a **half-yearly** paper on the term-1 chapters and an **annual** paper on the whole year. They go with *Think Like a Coder*, Class 8, and with the chapter assessments. All questions are new.

How they follow CBSE

- CBSE does not hold board examinations for Classes 3-8 and has not published a question-paper pattern for CT&AI.; Schools set their own papers. These papers follow what CBSE's CT&AI; handbooks describe: **competency-based** written tests with CT puzzles, with no 'write a program' questions and no computer needed.
- The question types are the ones CBSE uses in its competency-based papers: multiple choice, assertion-reason, short and long answers, and a case-based question.
- Questions are spread across four thinking levels (remember, understand, apply, analyse and create), with most marks for applying and analysing. The blueprints show the split.
- Chapters follow the order of CBSE's 2026-27 CT&AI; handbooks and the NCERT maths chapters, including the four AI chapters.
- The handbooks also assess Classes 6-8 through practical examinations, project presentations and reflective journals. The last section gives unplugged practical stations, a rubric and one way to combine the marks.

Paper design

Section	Question type	Questions	Marks
A	Multiple choice and assertion-reason	10	10 x 1 = 10
B	Very short answer	5	5 x 2 = 10
C	Short answer	5	5 x 3 = 15
D	Long answer	2	2 x 5 = 10
E	Case-based question	1	1 x 5 = 5
	Total · 2 hours	23	50

Using the papers

- Use the half-yearly paper in September and the annual paper in February-March, or as mock tests before the school's own exams.
- Print one paper per student; students answer on separate answer sheets.
- Read questions aloud to students who find reading hard: the paper tests thinking, not reading.
- Mark with the marking scheme; accept any correct method, and give part marks for correct steps.
- Use the blueprints to set your own papers in the same pattern.

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SAMPLE PAPER · HALF-YEARLY EXAMINATION

Class 8 · Computational Thinking and Artificial Intelligence

Syllabus: chapters 1-4 and AI 1-2 (term 1)

Time allowed: 2 hours **Maximum marks:** 50

Name: _____ **Section:** _____ **Roll no.:** _____

General instructions

1. This paper has 23 questions in 5 sections. All questions are compulsory.
2. Section A: multiple choice and assertion-reason, 10 questions of 1 mark each; Section B: very short answer, 5 questions of 2 marks each; Section C: short answer, 5 questions of 3 marks each; Section D: long answer, 2 questions of 5 marks each; Section E: case-based question, 1 question of 5 marks each.
3. Marks for each question are shown on the right.
4. Write your answers on the separate answer sheet. Draw tables and diagrams neatly where asked.
5. No calculator or computer is needed. Show your working: marks are given for correct steps.

SECTION A · Multiple choice and assertion-reason — 10 x 1 = 10 marks

1. Which number is a perfect cube? [1]

a) 64 **b)** 100 **c)** 144

2. The square root of 196 is ... [1]

a) 13 **b)** 14 **c)** 16

3. $2^3 \times 2^2 = \ldots$ [1]

a) 2^5 **b)** 2^6

c) 4^5

4. A coin is tossed 4 times. How many different results (such as HTHH) are possible? [1]

a) 8 **b)** 16 **c)** 4

5. The binary number 1010 is ... in decimal. [1]

a) 10 **b)** 12 **c)** 1010

6. A quadrilateral with two pairs of equal sides next to each other is a ... [1]

a) kite **b)** trapezium **c)** rectangle

7. The share of test predictions a model gets right is its ... [1]

a) accuracy **b)** deployment **c)** dataset

8. AI forecasts of air quality help to ... [1]

a) warn people on bad-air days **b)** count wildlife

c) translate languages

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9. **Assertion (A):** 72 is not a perfect square. [1]
Reason (R): 72 = 2 x 2 x 2 x 3 x 3, and its prime factors do not all pair up.

a) Both A and R are true, and R explains A.

b) Both A and R are true, but R does not explain A.

c) A is true, but R is false.

d) A is false, but R is true.

10. **Assertion (A):** Every problem needs an AI solution. [1]
Reason (R): AI learns patterns from data.

a) Both A and R are true, and R explains A.

b) Both A and R are true, but R does not explain A.

c) A is true, but R is false.

d) A is false, but R is true.

SECTION B · Very short answer — 5 x 2 = 10 marks

11. By what smallest number must 252 be divided to get a perfect square? What is the square root of the result? [2]

12. Simplify $(2^5 \times 2^3) \div 2^6$, and write 81 as a power of 3. [2]

13. Write 22 in base 3. [2]

14. One angle of a parallelogram is 65°. Find the other three angles. [2]

15. A no-code model mixes up paper and plastic in dim light. Give two ways to fix it. [2]

SECTION C · Short answer — 5 x 3 = 15 marks

16. Is 3375 a perfect cube? Use prime factors, and find its cube root if it is. [3]

17. A lock has 3 dials, each with the digits 0 to 9. How many codes are possible? How many with 4 dials? How many times harder is the 4-dial lock to guess? [3]

18. Change the binary number 101101 into decimal, and then into base 3. [3]

19. Use if-else rules on properties to name each shape: (a) only one pair of parallel sides; (b) two pairs of parallel sides, all sides equal, angles of 70° and 110°; (c) two pairs of parallel sides, all sides equal, all angles 90°. [3]

20. A model predicts 'rain' or 'no rain'. On 40 test days it is right on 34. Find its accuracy. The goal is 90%: should it be deployed? What next? [3]

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SECTION D · Long answer — 2 x 5 = 10 marks

21. Plan an AI project that predicts which canteen items will sell out each day.

(a) Define the problem in one sentence. [1]

(b) What data would you collect? [1]

(c) Give two cleaning steps. [1]

(d) How would you evaluate the model? [1]

(e) How would you deploy and keep checking it? [1]

22. Passwords and powers.

(a) How many 2-letter passwords can be made from the 26 small letters? [1]

(b) Write the number of 4-letter passwords as a power. [1]

(c) If the digits 0-9 are also allowed, write the number of 4-character passwords as a power. [1]

(d) Why does each extra character make a password much harder to guess? [1]

(e) Work out 2^{10}. Why do people say it is 'about a thousand'? [1]

SECTION E · Case-based question — 1 x 5 = 5 marks

23. An AI tool reads chest X-rays to help screen people for tuberculosis (TB) in villages without many specialists.

(a) Which AI domain is this? [1]

(b) What labelled data was it trained on? [1]

(c) Why must a doctor review its results? [1]

(d) Give one way it helps people in villages. [1]

(e) What could go wrong if all its training X-rays came from one hospital's machine? [1]

***** End of the question paper *****

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SAMPLE PAPER · ANNUAL EXAMINATION

Class 8 · Computational Thinking and Artificial Intelligence

Syllabus: all chapters, mostly chapters 5-7 and AI 3-4 (term 2)

Time allowed: 2 hours **Maximum marks:** 50

Name: _____ **Section:** _____ **Roll no.:** _____

General instructions

1. This paper has 23 questions in 5 sections. All questions are compulsory.
2. Section A: multiple choice and assertion-reason, 10 questions of 1 mark each; Section B: very short answer, 5 questions of 2 marks each; Section C: short answer, 5 questions of 3 marks each; Section D: long answer, 2 questions of 5 marks each; Section E: case-based question, 1 question of 5 marks each.
3. Marks for each question are shown on the right.
4. Write your answers on the separate answer sheet. Draw tables and diagrams neatly where asked.
5. No calculator or computer is needed. Show your working: marks are given for correct steps.

SECTION A · Multiple choice and assertion-reason — 10 x 1 = 10 marks

1. Which number is divisible by 4? [1]

a) 1,234 **b)** 3,516 **c)** 2,822

2. What is the remainder when 2,026 is divided by 9? [1]

a) 1 **b)** 2 **c)** 6

3. 99 x 101 = ... [1]

a) 9,999 **b)** 10,099 **c)** 9,899

4. $(a - b)^2$ = ... [1]

a) $a^2 - b^2$

b) $a^2 - 2ab + b^2$

c) $a^2 + b^2$

5. 15 : 25 in simplest form is ... [1]

a) 3 : 5 **b)** 5 : 3 **c)** 1 : 5

6. A group with far too many examples in the training data is ... [1]

a) over-represented **b)** under-represented

7. Checking a news story against trusted sources is called ... [1]

a) fact-checking **b)** phishing **c)** piracy

8. The angles of a quadrilateral add up to ... [1]

a) 180° **b)** 360° **c)** 540°

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9. **Assertion (A):** 8,316 is divisible by 11. [1]
Reason (R): Its alternating digit sum is 8 - 3 + 1 - 6 = 0.

a) Both A and R are true, and R explains A.

b) Both A and R are true, but R does not explain A.

c) A is true, but R is false.

d) A is false, but R is true.

10. **Assertion (A):** If 4 workers finish a job in 6 days, 8 workers finish it in 12 days. [1]
Reason (R): When more workers share a job, it takes fewer days (inverse proportion).

a) Both A and R are true, and R explains A.

b) Both A and R are true, but R does not explain A.

c) A is true, but R is false.

d) A is false, but R is true.

SECTION B · Very short answer — 5 x 2 = 10 marks

11. Test 5,148 for divisibility by 3, 4 and 9 without dividing. [2]

12. Use identities to work out 48^2 and 52 x 48. [2]

13. 6 kg of rice costs ₹390. Use the unitary method to find the cost of 10 kg. [2]

14. A dataset has 300 records of city students and 60 of village students. Which group is under-represented? Give one fix. [2]

15. A quiz app asks to use your camera and your contacts. Which would you allow? Why? [2]

SECTION C · Short answer — 5 x 3 = 15 marks

16. What day of the week is it 250 days after a Wednesday? Explain using remainders. [3]

17. Expand (x + 5)(x + 3) with an area model. Check your answer by putting x = 10. [3]

18. A map has the scale 1 : 50,000. How far is 6 cm on the map in real life? How long is a 4.5 km road on the map? [3]

19. A voice assistant understands men's voices well but often fails with women's and children's voices. Give the likely cause, how to test it, and a fix. [3]

20. An AI-made picture of a famous monument under floodwater is being shared widely. Name three checks before you share it. [3]

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SECTION D · Long answer — 2 x 5 = 10 marks

21. The identity $(a + b)(a - b) = a^2 - b^2$.

(a) Show why the identity is true by expanding the brackets. [2]

(b) Use it to work out 105 x 95. [1]

(c) Use it to work out $63^2 - 37^2$. [1]

(d) Why are identities like this useful for fast calculation? [1]

22. A school trip: the bus costs ₹12,000, shared equally by the students who go. Food costs ₹150 for each student.

(a) If 40 students go, how much does each pay for the bus? [1]

(b) If 60 students go, how much does each pay for the bus? [1]

(c) Is the bus share per student in direct or inverse proportion to the number of students? [1]

(d) What is the total food cost for 40 students, and for 60? [1]

(e) Is the total food cost in direct or inverse proportion to the number of students? [1]

SECTION E · Case-based question — 1 x 5 = 5 marks

23. A school uses AI to choose students for its science olympiad team. It learnt from five years of past choices, in which 80% of the students chosen were boys.

(a) What kind of bias is this? [1]

(b) How would you check whether it is fair? [1]

(c) How would you rebalance it? [1]

(d) Which feature should it not use? [1]

(e) Who should make the final choice, and why? [1]

***** End of the question paper *****

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Blueprints

Blueprint: Half-yearly examination

Syllabus: chapters 1-4 and AI 1-2 (term 1). Marks by chapter and section.

Chapter	A	B	C	D	E	Marks
1. A Square and a Cube	3	2	3			8
2. Power Play	2	2	3	5		12
3. A Story of Numbers	1	2	3			6
4. Quadrilaterals	1	2	3			6
AI 1. AI Project Lifecycle	2		3	5		10
AI 2. AI and Its Applications	1	2			5	8
Total	10	10	15	10	5	50

Marks by thinking level

Each question is counted at its main thinking level.

Level	What the student does	Marks	Share
Remember	recall a fact or term	2	4%
Understand	explain or interpret an idea	3	6%
Apply	use an idea or rule in a new problem	31	62%
Analyse and create	trace, debug, compare, design or justify	14	28%

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Blueprint: Annual examination

Syllabus: all chapters, mostly chapters 5-7 and AI 3-4 (term 2). Marks by chapter and section.

Chapter	A	B	C	D	E	Marks
4. Quadrilaterals	1					1
5. Number Play	3	2	3			8
6. We Distribute Yet Things Multiply	2	2	3	5		12
7. Proportional Reasoning	2	2	3	5		12
AI 3. Data and Fairness in AI	1	2	3		5	11
AI 4. Ethics and Responsible AI	1	2	3			6
Total	10	10	15	10	5	50

Marks by thinking level

Each question is counted at its main thinking level.

Level	What the student does	Marks	Share
Remember	recall a fact or term	2	4%
Understand	explain or interpret an idea	5	10%
Apply	use an idea or rule in a new problem	29	58%
Analyse and create	trace, debug, compare, design or justify	14	28%

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Marking schemes

Accept any correct method and any sensible wording. Give part marks for correct steps, even when the final answer has a slip.

Marking scheme: Half-yearly examination

1. a) 64
2. b) 14
3. a) 2^5
4. b) 16
5. a) 10
6. a) kite
7. a) accuracy
8. a) warn people on bad-air days
9. a) Both A and R are true, and R explains A.
10. d) A is false, but R is true. (Some problems need only fixed steps: automation.)
11. 252 = 2 x 2 x 3 x 3 x 7, so divide by 7. 36, whose square root is 6. (1 mark for 7, 1 mark for 6)
12. 2^2 = 4; 3^4 (1 mark for each)
13. 211 (2 x 9 + 1 x 3 + 1) (1 mark for the method, 1 mark for 211)
14. 115°, 65°, 115° (2 marks; 1 mark for 115°)
15. Add more photos of both classes taken in dim light; add more variety in backgrounds and angles; then retrain and test. (1 mark for each fix)
16. 3375 = (3 x 3 x 3) x (5 x 5 x 5): the factors make groups of three, so yes. Cube root = 3 x 5 = 15. (1 mark for the prime factors, 1 mark for the groups, 1 mark for 15)
17. 10^3 = 1,000; 10^4 = 10,000; 10 times (1 mark for each)
18. 45; 1200 in base 3 (1 x 27 + 2 x 9 + 0 + 0) (1 mark for 45, 2 marks for 1200 (1 mark for the method))
19. (a) trapezium (b) rhombus (c) square (1 mark for each)
20. 85%. No, it is below the goal. Collect more and cleaner data, refine the model and test again. (1 mark for 85%, 1 mark for the decision, 1 mark for the next step)
21. Open answer, for example: (a) predict which items will sell out so the canteen makes enough (b) past daily sales, day of the week, weather, events (c) remove duplicate days; fill or drop missing sales (d) test on days the model has not seen and measure accuracy (e) use it daily, check its predictions each week and retrain (1 mark for each stage)
22. (a) 26^2 = 676 (b) 26^4 (c) 36^4 (d) each extra character multiplies the number of passwords by 36, so guessing takes 36 times longer (e) 2^{10} = 1,024, which is just over a thousand (1 mark for each part)
23. (a) Computer Vision (b) many X-rays, each labelled 'TB' or 'no TB' by doctors (c) the AI can be wrong, and a doctor is responsible for the diagnosis (d) more people can be screened quickly and early (e) it may work badly on X-rays from other machines, so test it on varied images (1 mark for each part)

Marking scheme: Annual examination

1. b) 3,516
2. a) 1
3. a) 9,999
4. b) $a^2 - 2ab + b^2$
5. a) 3 : 5
6. a) over-represented
7. a) fact-checking
8. b) 360°
9. a) Both A and R are true, and R explains A.
10. d) A is false, but R is true. (8 workers take 3 days.)
11. Digit sum 18: divisible by 3 and 9. Last two digits 48: divisible by 4. (1 mark for 3 and 9, 1 mark for 4)
12. $(50 - 2)^2$ = 2,304; $50^2 - 2^2$ = 2,496 (1 mark for each)
13. ₹65 per kg, so ₹650 (1 mark for each step)
14. Village students. Collect more village records (or check the model on each group separately). (1 mark for each)
15. Neither: a quiz does not need them to work (1 mark for the decision, 1 mark for the reason)
16. 250 = 35 x 7 + 5, so the remainder is 5. 5 days after Wednesday is Monday. (1 mark for the remainder, 1 mark for Monday, 1 mark for the explanation)
17. $x^2 + 3x + 5x + 15 = x^2 + 8x + 15$. Check: 15 x 13 = 195 and 100 + 80 + 15 = 195. (1 mark for the four parts, 1 mark for the expansion, 1 mark for the check)
18. 6 x 50,000 = 3,00,000 cm = 3 km; 4.5 km = 4,50,000 cm, which is 9 cm on the map (1 mark for the method, 1 mark for each answer)
19. Cause: few women's and children's voices in the training data. Test: measure accuracy for each group separately. Fix: add more varied voices and retrain. (1 mark for each)
20. Look for the story on trusted news sites; check who posted it first and when; look for signs of editing or AI (odd details), or do a reverse image search. (1 mark for each check)
21. (a) $(a + b)(a - b) = a^2 - ab + ab - b^2 = a^2 - b^2$ (b) $100^2 - 5^2$ = 9,975 (c) (63 + 37)(63 - 37) = 100 x 26 = 2,600 (d) it swaps a hard multiplication for easy squares and one subtraction ((a) 2 marks (b) 1 mark (c) 1 mark (d) 1 mark)
22. (a) ₹300 (b) ₹200 (c) inverse (d) ₹6,000 and ₹9,000 (e) direct (1 mark for each part)
23. (a) historical bias (b) compare how often girls and boys with similar scores are chosen (c) add more examples of girls who did well, or train only on scores, and test again (d) gender (e) the teachers, because people stay accountable for decisions made with AI (1 mark for each part)

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Practical and project assessment

CBSE's Classes 6-8 handbooks assess CT&AI; through practical examinations, project presentations and reflective journals as well as written tests. CBSE has not fixed how the marks are split, so the school decides. One way to make a 100-mark annual result:

Component	Marks	Evidence
Written examination (this paper)	50	Annual sample paper
Practical examination (unplugged)	20	Two stations from the list below, 10 marks each
Projects (two 20-hour projects)	20	Project rubrics in the workbook; presentation and data sheets
Observation journal and reflection	10	Teacher Observation Journal levels from the teacher book

Practical examination stations

Each station takes about 15 minutes. Students work alone or in pairs with paper, cards or the classroom kit; the examiner watches, asks the two questions shown, and marks with the rubric. No computer is needed.

Station	Task	Examiner asks
1. Factor tiles (chapter 1)	Break three numbers into prime-factor tiles and decide whether each is a square, a cube, both or neither.	How do pairs and groups of three show squares and cubes?
2. Base converter (chapter 3)	Show one number in base 10, base 2 and base 3 with place-value cards.	How do you convert by repeated division? Why does binary suit computers?
3. Shape classifier (chapter 4)	Sort twelve quadrilateral cards with a chain of if-else questions, then write the rules.	Why must 'square' be checked before 'rectangle'? Is every square a rhombus?
4. Divisibility race (chapter 5)	Test six numbers for divisibility by 3, 4, 8, 9 and 11 without dividing.	Why does the digit-sum rule work for 9?
5. Clean the data (AI 1)	Find and fix the problems in a printed dataset (missing, duplicate, wrong and irrelevant values), then compute accuracy from a results table.	Why must data be cleaned before training? Is the accuracy good enough to deploy?
6. Fairness audit (AI 3)	Check a printed training set for over- and under-represented groups and plan how to rebalance it.	Which group is missing? How would you test the fixed model?

Rubric for each station (10 marks)

Criterion	Marks	Full marks when the student …
Understands the task	2	restates the goal and what a correct result looks like
Method	4	follows or designs clear, ordered steps; uses the right CT or AI idea
Result and checking	2	gets a correct result and checks or tests it
Explaining	2	answers the examiner's questions with reasons, using the right words

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